🧪 AI Assistant Builder Lab (Flowise + OpenRouter) – MASTER VERSION

🎯 Objective

Build a **real-world AI assistant** using:

- Flowise (visual pipeline builder)
- OpenRouter API (LLM integration)

You will:

- Design structured conversation flows
- Build multi-step assistants
- Integrate AI responses
- Deploy using shareable link or embed

📌 🧩 Choose ONE Project Only

1. 🛒 AI Shopping Assistant
2. 💬 Customer Support Bot (FAQ + AI Fallback)
3. 💼 Resume Screening Assistant
4. 🎓 Course Recommendation Assistant
5. 🧠 Multi-Step Decision Assistant
6. 📄 Document Q&A Assistant (Mini RAG)

⏱️ Time Limit: 120 Minutes

Task	Time
Understand use case	10 mins
Design flow	15 mins
Build in Flowise	40 mins

Testing	20 mins
Deployment	15 mins
Bonus	20 mins

🛠️ Common Setup (For ALL Projects)

Step 1: Open Flowise

- Open your Flowise instance

Step 2: Create New Chatflow

- Click **New Chatflow**
- Name your project

Step 3: Add Core Nodes

- Chat Input
- Prompt Template
- LLM (OpenRouter)
- Chat Output

🔧 Connect OpenRouter

- Add LLM node
- Select OpenRouter
- Paste API key
- Choose model (Mixtral / LLaMA etc.)

🛒 1. AI Shopping Assistant

🎯 Goal

Recommend products based on user preferences.

🔹 Flow Design

- Welcome → Category → Budget → Recommendation → End

🔹 Steps

1. Ask user category
2. Ask budget
3. Pass inputs to prompt
4. Generate recommendations using LLM
5. Return top 3 products

💡 Prompt Hint

“You are a shopping assistant. Recommend 3 products based on category and budget.”

💬 2. Customer Support Bot (FAQ + AI Fallback)

🎯 Goal

Answer FAQs and fallback to AI if unknown.

🔹 Flow Design

- Welcome → FAQ Options → Answer → Unknown → AI

🔹 Steps

1. Create FAQ responses
2. Add condition logic (if known → answer)
3. Add LLM fallback node
4. Handle unknown queries

💡 Prompt Hint

“If query not in FAQ, provide helpful support response.”

💼 3. Resume Screening Assistant

🎯 Goal

Evaluate candidate suitability.

🔹 Flow Design

- Ask skills → Ask experience → Evaluate → Output

🔹 Steps

1. Collect skills
2. Collect experience
3. Pass to LLM
4. Generate evaluation
5. Suggest fit level

💡 Prompt Hint

“Evaluate candidate fit for role and give structured feedback.”

🎓 4. Course Recommendation Assistant

🎯 Goal

Suggest courses based on interest.

🔹 Flow Design

- Interest → Level → Recommendation

🔹 Steps

1. Ask interest domain
2. Ask skill level
3. Generate recommendations
4. Suggest learning path

💡 Prompt Hint

“Suggest beginner/intermediate courses with clear progression.”

🧠 5. Multi-Step Decision Assistant

🎯 Goal

Guide users through decisions.

🔹 Flow Design

- Input → Condition → Branch → Output

🔹 Steps

1. Ask multiple inputs
2. Add branching logic
3. Use LLM for reasoning
4. Provide final recommendation

💡 Prompt Hint

“Guide user decision step-by-step with reasoning.”

📄 6. Document Q&A Assistant (Mini RAG)

🎯 Goal

Answer questions from uploaded document.

🔹 Flow Design

- Upload → Embed → Retrieve → Answer

🔹 Steps

1. Add Document Loader
2. Add Text Splitter
3. Add Vector Store
4. Connect Retriever
5. Connect to LLM
6. Answer queries

💡 Hint

- Use small document for speed
- Keep chunk size small

⭐ Bonus Activities (Optional but Recommended)

🔥 Add Knowledge Base

- Upload FAQs / documents
- Connect retriever

🔥 Hybrid Bot

- Combine rules + AI fallback

🔥 Improve Prompts

- Structured outputs
- Bullet responses

🔥 Deploy Frontend

Create index.html

```
<!DOCTYPE html>
<html>
<body>
<h1>My AI Assistant</h1>
<script src="YOUR_WIDGET_SCRIPT"></script>
</body>
</html>
```

🔥 Host Application

Use:

- GitHub Pages
- Netlify
- Vercel

✅ Expected Output

- Working assistant
- Structured flow (min 5 steps)
- OpenRouter integrated
- Shareable link or hosted page

🏁 Submission Requirements

- Project title
- Flowise link
- Short description
- Screenshot

🎓 Evaluation Criteria

Criteria	Marks
Flow design	20
Functionality	25
Prompt quality	20
Use-case clarity	15
UI / UX	10
Bonus	10

🚀 Final Note

This is not just a chatbot task.
👉 Think like you are building a **real AI product**.